



EUSTRESS ENGINE

SOURCE-AVAILABLE · PHYSICS-AS-CODE · BUILT FOR INVENTORS

ALPHA PILOT PROGRAM · 2026

Center for Innovation

University of Arizona · Wildcat Cohort

Simulate anything. Validate faster. Ship with conviction.

Eustress is a real-time, file-system-first simulation engine where founders model their product in 3D, run it under real physics, and converse with an AI co-engineer that can read the scene, edit it, and explain results.

One API key. Real outcomes. No black box.

⚡ WORKSHOP · TALK TO EUSTRESS

Claude-powered agent with **full access to the running simulation** via MCP tools.
Domain modes (Simulation, Manufacturing, Fabrication, Supply Chain, Finance) gate the right tools for the right job.
Bring your own API key — the engine handles the rest.

🔬 REAL PHYSICS, NOT EFFECTS

Shipped laws: **Nernst, Butler-Volmer, Tafel, ohmic IR drop, Monroe-Newman dendrite risk, thermodynamics, conservation, classical mechanics.**
Time compression up to **1 year per second** for cycle-life and aging studies.

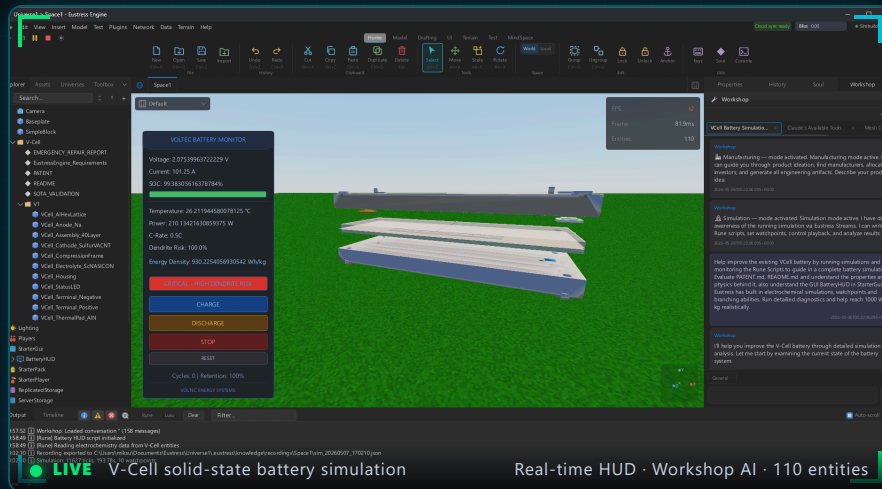
📡 EUSTRESS STREAMS · BRING YOUR DATA

Pipe live signals from **sensors, MQTT brokers, databases, REST APIs, or shared memory** directly into the 3D scene.
Every entity becomes a real-time observable.

622k msg/s · 411 μs · 20.4M shared-memory

📜 RUNE SCRIPTING · ZERO-COPY ECS

Type-safe, sandboxed scripting with **direct ECS memory access** — about 50× faster than Lua FFI.
VM pooling executes hot scripts in microseconds. Domain knowledge lives in code, not config.



STEM

Batteries, EVs, fuel cells, solar arrays, grid storage



CIVIL
Structural loads, thermal envelopes, smart-city flow



AEROSPACE
Orbits, propulsion, sensor packs, satellites



LIFE SCIENCES
Diffusion, kinetics, reaction networks, dosing



BIOTECH
Bioreactors, assays, microfluidics, fermentation



ROBOTICS
Control loops, kinematics, perception, swarms



MANUFACTURING
BOM, supplier risk, cost rollup, line balancing



AGTECH
IoT sensors, climate, irrigation, yield models



MEDDEVICE
Mechanical, fluidic, thermal, regulatory artifacts

01

Download

Free to download, source-available.
Windows, macOS, Linux. Vulkan / DX12 ready.

02

Drop your API key

Workshop activates. Talk to Eustress. It builds, edits, and explains.

03

Stream data in

Sensors, databases, telemetry. Bind any source to any entity.

04

Iterate at compression

Hours of behavior in seconds. Watchpoints, breakpoints, recordings.

★ WHY THIS IS A GAME-CHANGER FOR FOUNDERS

Most simulation tooling is locked behind seat licenses, opaque file formats, and consultant hours. Eustress flips that: **your project is a folder of plain-text TOML** that lives in git, your physics laws are **auditable Rust source**, and your AI co-engineer can read and edit every part of it. **Iterate in minutes, not months.** Bring a thesis to a defensible prototype before you ever write a check for hardware. When you pitch, you walk in with a real, runnable model — not a slide.

✓ PILOT PROGRAM 100% FREE

Wildcat cohort participation is **free for the alpha window**. No license fees, no seat minimums, no lock-in. You keep your data and your repo. In return we ask only that you **build loudly**: post in the Slack, file the rough edges you find, and let us ship a reply. Founders who help shape the engine in alpha get permanent recognition in the project and a fast lane on future feature requests.

Wildcat Founders: claim your alpha seat.

The Center for Innovation cohort gets **direct engineering access**, dedicated Slack channel, and one-on-one onboarding for your first simulation. Help shape the source-available platform of the future for inventors. Built in the open, and every line of it is yours to read, fork and build on.

— We cannot wait to see what you build.

▼ Download the Engine
eustress.dev/download

Join the Slack
eustressdev.slack.com

★ Learn the Engine
eustress.dev

ALPHA

Source-available (PolyForm Shield 1.0.0) · Read, fork and build on it · Git-diffable projects · No vendor lock-in

eustress.dev · #BearDown · #UofA · #UACI